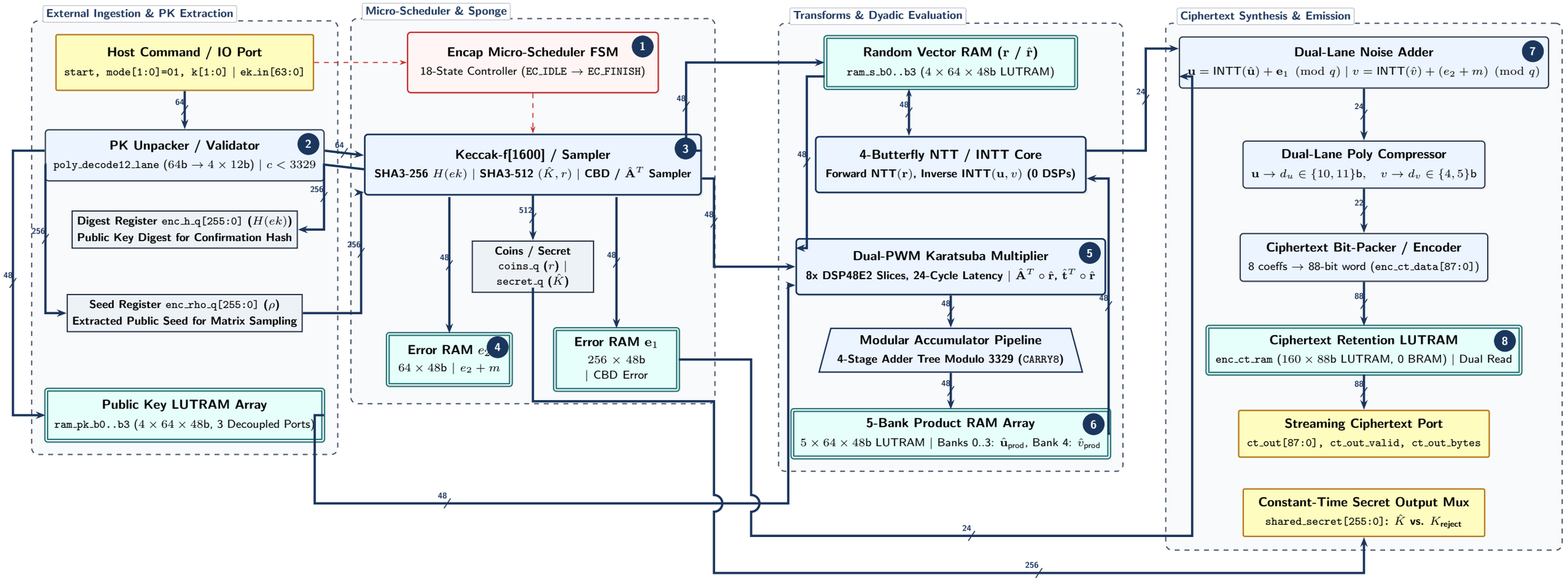
ML-KEM HPKA: Encapsulation Datapath & Micro-Scheduler Architecture

Subsystem 02 Microarchitecture — RTL Modules: `mlkem_encap_datapath.svh`, `mlkem_keygen_top.sv`



ID	Component / Signal	Width / Format	Description
1	Encap Micro-Scheduler	18-State FSM	Coordinates the 18-state encapsulation flow (`EC_0` to `EC_17`), polynomial loop counters, and coprocessor handshaking.
2	Public Key Decoder	64b → 48b Stream	Deserializes 64-bit ek stream into four 12-bit coefficients per clock, verifies $c < 3329$, and extracts seed ρ.
3	Keccak Sponge Core	64b Lane Sponge	Computes SHA3-256 $H(ek)$, SHA3-512 $(\hat{K}, r)$, CBD noise $(\mathbf{r}, \mathbf{e}_1, e_2)$, and matrix rejection sampling $\hat{\mathbf{A}}^T$.
4	Pipelined Message Adder	64 × 48b LUTRAM	3-stage adder pipeline computing $e_2 + \text{Decompress}_1(m) \cdot 1665 \pmod q$ concurrently with CBD sampling beats.
5	Pointwise Multiplier	8x DSP48E2 Slices	Pipelined dual-channel Karatsuba multiplier executing $\hat{\mathbf{A}}^T \circ \hat{\mathbf{r}}$ and $\hat{\mathbf{t}}^T \circ \hat{\mathbf{r}}$ with fixed 24-cycle latency.
6	Product Accumulator RAM	5 × 64 × 48b LUTRAM	5-bank zero-BRAM storage buffering vector products $\hat{\mathbf{u}}_{\text{prod}}$ (banks 0–3) and scalar product $\hat{v}_{\text{prod}}$ (bank 4).
7	Dual-Lane Noise Adder	2 × 12b Coeffs	Computes $\mathbf{u} = \text{INTT}(\hat{\mathbf{u}}) + \mathbf{e}_1 \pmod q$ and $v = \text{INTT}(\hat{v}) + (e_2 + m) \pmod q$ before poly compression.
8	Ciphertext Retention RAM	160 × 88b LUTRAM	Holds compressed ciphertext with dual read ports for decapsulation lockstep comparison and host readout.